

System Requirements Specification

This system is a **multi-branch restaurant management system** designed to manage restaurant operations including user access, branch management, menu control, order processing, payments, and sales monitoring.

The system supports two roles:

- **Admin**
 - **Cashier**
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1) Authentication & Authorization

The system must allow users to log in securely based on their role.

Admin

- Admin logs in using:
 - username
 - password
 - role
- Admin is **not required to belong to a specific branch**
- Admin has access to system-wide monitoring and management

Cashier

- Cashier logs in using:
 - username
 - password
 - role
 - branch

- Cashier must belong to exactly one branch
 - Cashier can only access operations related to their assigned branch
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2) User Management

This module is accessible only by admins.

Functionalities

- Create new users
- Assign user role:
 - admin
 - cashier
- Assign branch for cashier users
- Edit personal information of existing users
- Deactivate existing users without deleting historical data

Business Rules

- A cashier must belong to one branch
 - An admin may have no branch assignment
 - A branch can have multiple cashier users
 - Deactivation must not affect previous orders or payments
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3) Branch Management

This module allows admins to manage restaurant branches and their tables.

Functionalities

- Create new branches
- Add tables to a branch
- Delete tables from a branch

Business Rules

- Each branch can have multiple tables
 - Each table belongs to exactly one branch
 - A table can have only one active order at a time
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4) Menu Management

The system must support branch-specific menus and pricing.

Categories

- Admin can create categories
- Admin can delete categories
- Each category can contain multiple products

Products

- Admin can create products
- Admin can delete products
- Each product includes:
 - name
 - category
 - image
- Each product belongs to exactly one category

Menu Items

Each product with a specific size is considered a separate menu item.

Business Rules

- Each menu item belongs to exactly one branch
- The same product may exist in multiple branches

- The same product may have different prices across branches
 - Price is stored at the menu item level
 - Size is stored at the menu item level
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5) Extras Management

The system must support optional extras that can be added to menu items.

Business Rules

- A menu item may support multiple extras
 - An extra may be available for multiple menu items
 - The price of an extra may vary depending on the menu item
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6) Order Management

Cashiers are responsible for creating and managing customer orders.

All orders are stored in a single order structure.

Supported Order Types

- dine-in
- takeaway
- external delivery

External Delivery Destinations

For external delivery orders, destination can be:

- Attara
- Falafel

For other order types, destination remains empty.

Business Rules

- Dine-in orders must be linked to a table

- Takeaway orders do not require a table
 - External delivery orders do not require a table
 - A table can only have one active order at a time
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7) Order Items

Each order may contain multiple order items.

Important Business Rule

The item price must be stored as a snapshot during ordering so historical orders remain accurate even if menu prices change later.

8) Order Item Extras

Customers may select extras for specific ordered items.

Business Rules

- One order item may contain multiple extras
 - Each selected extra must be stored separately
 - Historical prices must be preserved
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9) Payment Management

Cashiers can complete order checkout for standard orders. The system also tracks Accounts Receivable for third-party external deliveries.

Supported Payment Methods

- Cash
- Card / Visa
- Third Party (Used for external delivery partners like Attara and Falafel)

Business Rules

- Each order has a maximum of one payment (split payments are not supported).
 - Each payment belongs to exactly one order.
 - Payments for internal orders (Dine-in / Takeaway) must be linked to the specific Cashier who processed the transaction.
 - Payments for external deliveries (Attara / Falafel) must use the "Third Party" payment method. These transactions do not require a Cashier and act as an Accounts Receivable record until the third party settles their balance.
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10) Order Audit Trail

The system must maintain a full audit trail for order activities.

This table is dedicated to **historical tracking only**.

Purpose

Track what happened to the order after creation.

Supported Event Types

- created
 - item added
 - item removed
 - closed
 - cancelled
 - Deleted
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11) Monitoring & Reporting

Cashier Access

Cashiers can view:

- today's order history for their branch

- total revenue for their branch

Admin Access

Admins can view:

- all branches data
- orders per branch
- external delivery orders
- total sales summary
- dashboard analytics
- revenue reports